

Imaging the Plastics on the SiPM Wall → a Geometry Calibration

n_TOF EAR2 X17 — run 224404, MIP coincidences + Geant4 geometry

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analysis and slides prepared with Claude (Anthropic AI assistant)

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Idea: no fitted energy scale needed — use the geometry

- ▶ Take the wall–plastic coincidences, **assume they are MIPs**, and **assume they come from the source** (He-3 target at the origin).
- ▶ Geometry from the Geant4 sim (mx17_geom.py): bars run *along* the beam (+Y), wall at $R \approx 33$ cm, plastics behind at $R \approx 43$ cm.
- ▶ A source track hits the wall at angle θ , so a MIP deposits $E = \frac{dE}{dx} \cdot \frac{t_{\text{scint}}}{\cos \theta}$ in the 3 mm bar.
- ▶ Two consequences:
 1. each plastic *projects* onto the wall $\Rightarrow$ imaging;
 2. the MIP peak must scale as $1/\cos \theta \Rightarrow$ a calibration.

$$\cos \theta = R_{\text{wall}} / \sqrt{R_{\text{wall}}^2 + u^2 + v^2}$$

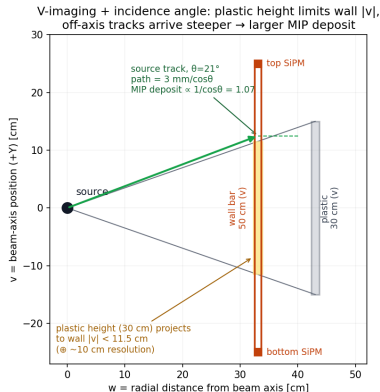
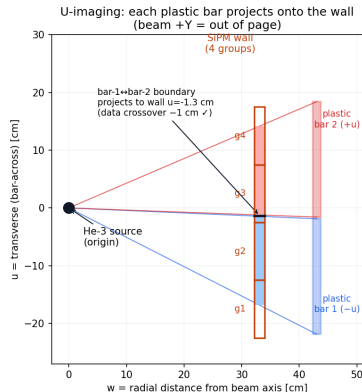
plastic $\rightarrow$ wall projection scale

$$R_{\text{wall}}/R_{\text{plastic}} = 0.77$$

$$\Rightarrow \text{geo-mean MIP peak} \propto \text{gain} \times \frac{1}{\cos \theta}$$

Geometry: the two projections and the coordinates

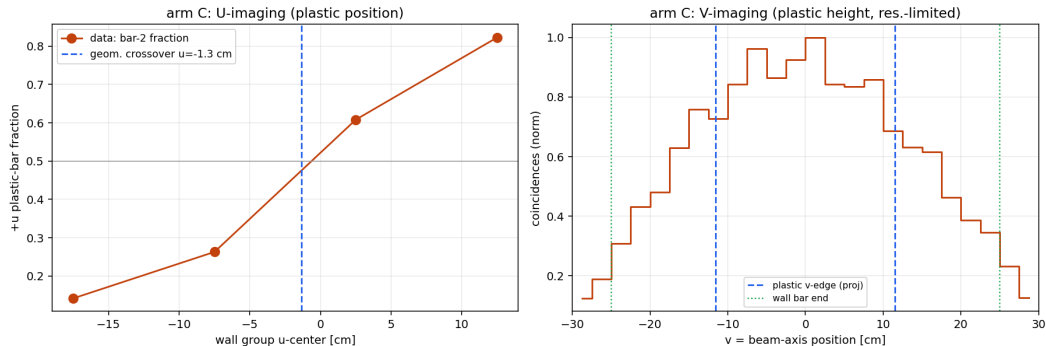
MX17 arm C: source $\rightarrow$ wall $\rightarrow$ plastic geometry ($R_{\text{wall}}=33$ cm, $R_{\text{plastic}}=43$ cm, projection $\times 0.77$)



u = transverse (bar-across), v = beam axis ($+Y$, bar length, read top/bottom), w = radial. *Left:* rays from the source through each plastic bar illuminate different wall groups (blue bar-1 $\rightarrow$ g1,2; red bar-2 $\rightarrow$ g3,4). *Right:* the plastic height limits the wall $|v|$, and an off-axis track arrives at angle $\theta \Rightarrow$ longer path $\Rightarrow$ larger MIP deposit.

Imaging: the wall sees the plastics from the source

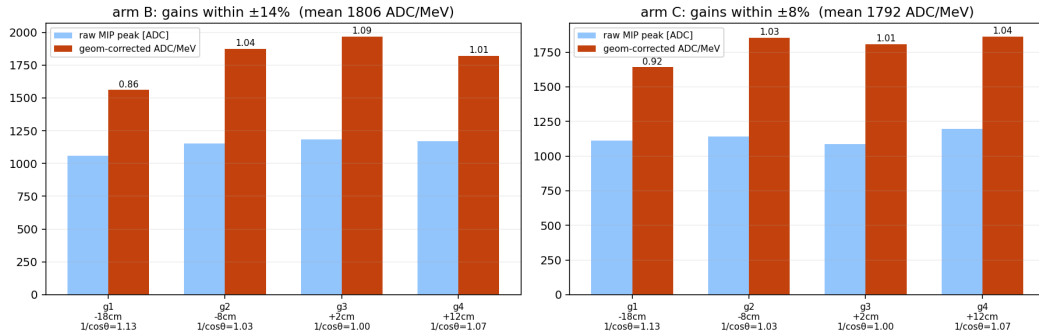
run224404: imaging the plastics onto the SiPM wall from the source



Left (U): fraction of coincidences tagged by the $+u$ plastic bar climbs monotonically across the wall groups, crossing 50% at $u \approx -1$ cm — exactly the source-projected plastic-bar boundary (-1.3 cm). *The tracks really do come from the source.* **Right (V):** coincidences centre at $v = 0$ and fall off near the projected plastic height (± 11.5 cm); the soft edges are the ~ 10 cm timing/amplitude resolution.

Calibration: correct the MIP peak by $1/\cos\theta$

run224404: geometry-corrected wall calibration (number above bar = relative SiPM gain)



Raw MIP peak (blue) is nearly flat across the wall; dividing by the geometric deposit $E = 0.6 \text{ MeV} / \cos\theta$ gives **ADC/MeV** (orange). Per-group **relative SiPM gain** (numbers) sits within $\pm 8\text{--}14\%$; group 1 (far $-u$ edge) is $\sim 10\%$ low on both arms. Mean $\approx 1800 \text{ ADC/MeV}$.

What this gives — and its limits

- ▶ **Geometry & source confirmed** by the U-imaging (bar-fraction gradient matches the projected plastic boundary on both arms).
- ▶ **Relative per-group gain map** (equalisation) from MIP peak / $(1/\cos\theta)$ — the robust deliverable, $\pm 8\text{--}14\%$.
- ▶ **Absolute ADC/MeV** ≈ 1800 *under the MIP assumption* using $dE/dx = 2.0 \text{ MeV/cm} \times 0.3 \text{ cm}$. Caveat: this uses the *mean* dE/dx ; the Landau most-probable deposit is $\sim 30\%$ lower, an $O(1)$ scale offset on the absolute number (relative gains unaffected).
- ▶ **V-imaging is resolution-limited** ($\sigma_v \sim 10 \text{ cm}$ vs $\pm 11.5 \text{ cm}$ plastic projection): the geometric $1/\cos\theta$ trend *within* a group ($\sim 5\%$) cannot be resolved — the calibration rests on the across-group correction + the validated geometry.
- ▶ **Per-group only**: 4 bars are ganged per group, so this is a u -map at group granularity (4 points), v at $\sim 10 \text{ cm}$.
- ▶ Next: a triggered / known-position sample (or LS readout) pins v_{eff} , λ , and the absolute MeV scale, upgrading this to a per-bar calibration.